

Experiment 7: Arrays

Objective: To learn how to declare, initialize, and manipulate arrays.

Questions:

1. Write a C program to find the largest element in an array of size 10.

Sample Input:

Enter Array Elements: 2, 78, 7, 8, 10, 67, 44, 33, 29, 5

Sample Output:

78

2. Write a C program to find the sum and average of all elements in an array.

Sample Input:

Enter array elements: 1 2 3 4 5

Sample Output:

Sum of array elements: 15

Average of array elements: 3.00

3. Write a C program to reverse the elements of an array.

Sample Input:

Enter array elements: 10 20 30 40 50

Sample Output:

Reversed array: 50 40 30 20 10

4. Write a C program to count the frequency of each element in an array.

Sample Input:

Enter Array Elements: 2, 78, 78, 10, 67, 2, 33, 29, 5

Sample Output:

Frequency:

2 - 2

78 - 2

10 - 1

67 - 1

33 - 1

29 - 1

5 - 1

5. Write a program to find the second largest element in an array.

Sample Input:

Enter array: 20 10 50 30 40

Sample Output:

The second largest element is 40.

6. Write a C program to count even and odd elements in an array using pointers.

Sample Input:

Enter Array Elements: 2, 78, 7, 8, 10, 67, 44, 33, 29, 5

Sample Output:

Even Elements: 2, 78, 8, 10, 44

Odd Elements: 7, 67, 33, 29

7. Write a function that takes a string and a character, and returns the first position where the character occurs.

Sample Input:

Enter String: "Array", Enter character: 'r'

Sample Output: 2

8. Write a C program to enter the size of 2 matrices. Check whether matrix multiplication is possible or not. If yes, provide output. Use functions to reduce redundancy.

Sample Input:

For Array 1:

Enter rows: 2

Enter Columns: 2

For Array 2:

Enter rows: 2

Enter Columns: 2

For Array 1:

Enter Elements: 2 2 2 2

For Array 2:

Enter Elements: 2 2 2 2

Sample Output:

8 8 8 8